

Qns	Answer	Marks
7(a)(i)	$E = \frac{hc}{\lambda}$ $= \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{350 \times 10^{-9}}$ $E \approx 5.68 \times 10^{-19} \text{ J} = 3.55 \text{ eV}$ photon energy less than work function photoelectrons will not be emitted.	M1 A1
7(a)(ii)	the <u>energy of photon</u> remains the same, as a <u>single photon</u> can only interact with (and pass on its energy to) a single electron, answer in (a) is <u>unaffected</u>.	B1 B1
7(b)	$Pt = Nh\nu$ $N = \frac{P\lambda}{t \quad hc}$ $= \frac{0.50 \times 350 \times 10^{-9}}{6.63 \times 10^{-34} \times 3.0 \times 10^8}$ $= 8.80 \times 10^{17} \text{ s}^{-1}$	A1

Qns	Answer	Marks
7(c)(i)	$p = \frac{h}{\lambda}$ $p = \frac{h}{\lambda}$ $= \frac{6.63 \times 10^{-34}}{350 \times 10^{-9}}$ $= 1.89 \times 10^{-27} \text{ kg m s}^{-1}$ Since photons are absorbed by the metal surface, their resulting momentum is 0 after impact. By N2L, Force on photons by surface $= \frac{N \Delta p}{t}$ $= 8.80 \times 10^{17} \times [0 - (1.89 \times 10^{-27})]$ $= -1.66 \times 10^{-9} \text{ N}$ By N3L, force on surface by photons $= 1.66 \times 10^{-9} \text{ N}$ Hence, pressure on surface $P = \frac{1.66 \times 10^{-9}}{4.0 \times 10^{-7}}$ $= 4.15 \times 10^{-3} \text{ Pa}$	C1 (N2L) M1 (N3L) A1 (P = F/A)
7(c)(ii)	reflected photons have (non-zero) momentum so the <u>change in momentum of photons is higher</u> , hence force and pressure on surface is higher	B1
8(a)(i)	when two or more waves of the same type meet and overlap	B1